

THE OPEN SOURCE AUDIT

A Capstone Project Report

Course: Open Source Software | Units 1–5 | Max Marks: 100

Software Chosen:

**Firefox — Open Source Web Browser | Licence: Mozilla Public
Licence 2.0**

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1. Introduction

This report is a structured audit of one of the most well-known and most used open-source web browsers around the world. Unlike other browsers developed by companies such as Google, Apple, or Microsoft, Mozilla Firefox is not a product developed to represent the interests of a corporation and its advertisers. Mozilla Firefox is a product developed to represent its users. The Mozilla Foundation and its wholly owned subsidiary, Mozilla Corporation, are a non-profit organisation whose stated mission is to ensure the internet remains open, accessible, and safe for everyone.

Firefox occupies a special position in the history of open source. It was created out of the ashes of Netscape Navigator, once the preeminent browser before it was devastated by the anticompetitive tactics of Microsoft in the 1990s. To understand Firefox is to understand one of the most important struggles in the history of the Internet, and how open source became a way to fight back.

This report explores the Firefox browser as an open-source artefact under four sections for the Open Source Software course units 1-5. Five shell scripts developed using Kali Linux are included in the GitHub repository with this report.

Part A — Origin and Philosophy (Units 1 & 2)

A1. The Problem Firefox Was Created to Solve

In order to understand Firefox, it is important to take a brief look back into history, specifically at the events of 1995, known as the browser wars. Netscape Navigator was the first commercially successful browser. It was created by Netscape Communications Corporation. It was the first browser to introduce features that would define the early web, including JavaScript, cookies, and SSL encryption. At one time, Netscape Navigator dominated the browser market, having more than 80% of all browser users. The web, in a sense, was created for and within Netscape.

Microsoft saw that if Netscape controlled the browser, Netscape would control the Internet standards and would therefore be able to set Internet standards to benefit their own computer platform. In response to this threat, Microsoft gave Internet Explorer away for free with their Windows computer system. This strategy, which was subsequently deemed anticompetitive by the US federal courts, was extremely successful for Microsoft. By 2001, Internet Explorer's market share had increased to over 90%, and Netscape was dying.

However, in 1998, in a daring move, as the Netscape browser was on the verge of extinction, the organization made a historic decision: it made the source code of the Netscape browser freely available. This led to the development of a new project called Mozilla, which maintained and enhanced the source code of the Netscape Communicator browser. The term 'Mozilla' was coined as an acronym for 'Mosaic killer,' a term used for the Netscape browser, which was originally inspired by Mosaic, and Godzilla, a monster from a popular movie series.

Mozilla's aim was to recreate a free, open-source browser that could not be owned or killed by anyone or any organization.

The original Mozilla Suite was a large and slow product. In 2002, two developers within the Mozilla group, Dave Hyatt and Blake Ross, decided to create a new browser product called Phoenix, later renamed Firebird and finally Firefox in 2004. The plan was simple: take only the browser portion of the Mozilla code, make it fast, make it lightweight, and make it the best browser out there. On November 9, 2004, Firefox 1.0 was released, including a full-page ad in the New York Times paid for by donations. In one year, it was downloaded over 100 million times. It was one of the most successful product launches in history.

Mozilla Foundation was set up in 2003 as a non-profit organization that would be responsible for the intellectual property rights of Firefox, ensuring that it would remain a free and open project even after the acquisition of Netscape by AOL, which then shut down Netscape. The mission for which Mozilla was created still holds today: a free, open, and interoperable internet, free from monopoly control. Firefox was created because the alternative was a world where a single company controlled how everyone on the planet used the internet.

A2. The Licence — What MPL 2.0 Actually Says

Firefox is covered by the Mozilla Public Licence version 2.0 (MPL 2.0). This licence is a "middle ground" between the permissive licences (like MIT and BSD licences) and the strong copyleft licences (like GPL v2). To understand this "middle ground", we need to look at the four freedoms of free software as defined by the GNU Project.

The four freedoms are: (0) the freedom to run the program for any purpose; (1) the freedom to study and modify the source code; (2) the freedom to redistribute copies; and (3) the freedom to distribute modified versions. MPL 2.0 licenses all four freedoms. You can download the source code of Firefox from hg.mozilla.org

or GitHub and read every single line of it. You can modify it, run it, and share it. There are no restrictions on purpose or identity.

The main feature of MPL 2.0 is its file-level copyleft. Unlike GPL v2, where the entire modified software has to be open source, with MPL 2.0, only individual files need to be under an open source license. This means if you want to modify an MPL-licensed file, then that modified file has to be under MPL 2.0. However, with MPL 2.0, you can mix and match MPL-licensed files with proprietary files. This makes MPL 2.0 more business-friendly than GPL while still keeping the modified source of the Firefox codebase under an open source license.

Dimension	MPL 2.0 (Firefox's Licence)	GPL v2 (Linux / Git)
Copyleft scope	File-level — only modified MPL files must stay open	Project-level — entire modified work must be GPL
Combining with proprietary code	Permitted — proprietary files can be added alongside	Not permitted — all combined code must be GPL
Commercial use	Fully permitted with no source-sharing for new files	Permitted, but modified distributions must share source
Patent grants	Explicit patent grant from contributors — stronger protection	Implied protections only
Used by	Firefox, Thunderbird, LibreOffice	Linux kernel, Git, WordPress, GCC

A company can take Firefox source code and modify it to build a proprietary browser, but it must be noted that it must be licensed under MPL 2.0. This is exactly what has been done. Several browsers have been built on Firefox's rendering engine, called Gecko, or on the broader Mozilla code base. These include Pale Moon, Waterfox, and Basilisk. The file-level copyleft ensures that any enhancements to the core engine are fed back into the community. The flexibility of the license allows companies to build proprietary features around it.

A good example of this difference between 'free as in beer' and 'free as in freedom' can be given with the help of the history of Firefox. Internet Explorer was free as in beer because Microsoft was providing it for free, but it was not free as in freedom because, unlike Firefox, its source code was not available for viewing, changing, or auditing for safe use with the user's information. Firefox, on the other hand, is free as in beer because it can be downloaded for free, and free as in freedom because its source code can be viewed, modified, and redistributed as per the MPL 2.0 license.

If you are building a browser or a major user-facing application, MPL 2.0 is a very strong license for this situation. It will prevent the underlying codebase from being improved in secret by competitors, but will allow commercial software to be built on top of it. If you are building a command-line tool or library where widespread adoption is the primary goal, MIT or PSF would be a better choice. If you are building infrastructure that should never be closed and owned by a single entity, GPL v2 is the strongest license.

A3. The Ethics of Open Source

Should all software be open source? Firefox brings this question into sharper focus than almost any other software. A browser is not a special-purpose tool, but rather the means by which billions of people access information, do business, take care of their health, and communicate with each other. It knows what news sites you visit, what political opinions you seek, what diseases you research, and what products you are considering. In the hands of a company with a financial interest in advertising, it is extremely valuable and even menacing.

The ethical case for open-source browsers is unusually strong. When Chrome is the dominant browser, and Google is an advertising firm, a conflict of interest exists between the users of Chrome and the corporation that makes Chrome. An open-source browser, a browser with code that can be understood and verified by unbiased researchers, is not merely a technical preference; it's a democratic imperative. The very existence of Firefox is an argument that the system through which people interact with information should not be controlled by a corporation that benefits from knowledge of what information they are interacting with.

However, the counterargument, that browser development for profit results in a faster, more polished product, has some merit. The speed improvements that Chrome has developed between 2008 and 2015 far outstrip Firefox's improvements over a period in which Mozilla was struggling with technical debt. The advantage of a large corporation (Google had hundreds of engineers working on Chrome) is that they will always get a result that a nonprofit cannot hope for. The ethical conflict is very real: a browser that is free but technically inferior is not as useful for the user as a proprietary browser that is faster but spies on them.

Is it ethical to make money off of open source labor? The relationship of Mozilla to moneyed interests is a complex one. The main source of revenue for Mozilla Corporation has been its relationship with Google, whereby Google pays hundreds of millions of dollars a year to Mozilla to be the default search engine of Firefox. This means that a non-profit browser funded by the world's largest advertising firm is the main source of competition for the same firm's own browser. The relationship is not unethical, as Mozilla uses the money to fund its browser and internet health projects, but it does suggest that the relationship between being "open source" and being "commercially involved" is a nuanced one.

"Open source is an enabler, not a destroyer, of innovation in the browser space." The KHTML rendering engine, developed by the KDE community, was used by Apple to develop the WebKit browser, which was again used by Google to develop Blink, which is used by all major browsers, including Chrome, Edge, and Firefox, among others. Similarly, Mozilla's Gecko rendering engine, which is completely open source, has enabled innovation in standards-based technologies, including CSS, JavaScript, and privacy-related technologies like Enhanced Tracking Protection. A proprietary browser ecosystem would have meant slower, less interoperable standards that would have reflected the business interests of individual companies rather than the collaborative process of standards bodies.

Part B — Linux Footprint on Kali Linux (Unit 2)

Installation

Firefox is available in Kali Linux's official APT repository as the Firefox ESR (Extended Support Release) package, which receives security updates for an extended period. Installation:

```
sudo apt update && sudo apt install firefox-esr -y # Install Firefox ESR
firefox-esr --version # Confirm installation and
version
```

The standard (non-ESR) Firefox release can be downloaded as a tarball from mozilla.org for users who want the latest features without waiting for APT updates:

```
wget https://download.mozilla.org/?product=firefox-latest-ssl&os=linux64
tar -xjf firefox-*.tar.bz2
./firefox/firefox & # Run directly
```

Key Directories

Path	Purpose / Contents
/usr/bin/firefox-esr	Main Firefox executable — symlink to the actual binary
/usr/lib/firefox-esr/	Core Firefox application files, libraries, and assets
/usr/share/applications/	Desktop entry file (firefox-esr.desktop) for GUI integration
/usr/share/doc/firefox-esr/	Documentation, changelog, copyright file
~/.mozilla/firefox/	User profile directory — bookmarks, history, passwords, extensions
~/.cache/mozilla/firefox/	Browser cache — temporary downloaded web content
/etc/firefox/	System-wide Firefox policies and configuration (enterprise use)

User, Group and Security

Firefox runs as the currently logged in user, it does not need or use a special system user account for normal desktop usage. The binary and application files under `/usr/lib/firefox-esr/` are owned by root and have normal permissions, so it is not possible for any normal user to change the browser executable. The user-specific data like bookmarks, extensions, passwords, and history are stored in a special subdirectory under `~/.mozilla/firefox/` owned and accessible only by the user.

Firefox runs as the currently logged-in user — it does not require or use a separate system user for standard desktop use. The binary and application files under `/usr/lib/firefox-esr/` are owned by root with standard permissions, so no ordinary user can modify the installed browser binary. User-specific information such as bookmarks, extensions, passwords, and browsing history are kept under the user's `~/.mozilla/firefox/` profile directory, which is only accessible by the user.

This separation also has significant security implications, as the Firefox profile directory holds sensitive information, including saved password information (encrypted) and cookies. In the event of a Firefox compromise from a malicious webpage or extension, the attacker would have access to the user's profile information, but not system information, unless the system is further misconfigured. Running Firefox as the root user, which the previous version of Kali Linux did by default (to the frustration of security researchers

everywhere), is a dangerous practice because it violates this separation. Modern versions of Kali Linux strongly discourage running the desktop as the root user for this reason.

Service Management

Firefox is a desktop application, not a system service, and does not have a systemd unit file for standard use. It is launched and terminated by the user. On Kali Linux:

```
firefox-esr &           # Launch Firefox in background
pkill firefox-esr       # Terminate all Firefox processes
pgrep -la firefox       # List running Firefox processes with PIDs
```

In automated or headless testing environments, Firefox can be run without a GUI using its headless mode:

```
firefox-esr --headless --screenshot output.png https://example.com # Screenshot a
page
firefox-esr --headless --print-to-pdf output.pdf https://example.com # Save page as
PDF
```

Update Model

Firefox on Kali Linux receives security updates through APT for the ESR edition. Mozilla also ships updates through its own built-in update mechanism for non-APT installations. When a critical vulnerability is discovered — such as the zero-day CVE-2023-4863 (a heap buffer overflow in the WebP image library used by Firefox) — Mozilla typically releases a patch within 24 to 48 hours. This rapid response is only possible because the source code is developed and reviewed publicly on Phabricator (Mozilla's code review platform) and GitHub, allowing security researchers worldwide to contribute to and review fixes.

```
sudo apt update && sudo apt upgrade firefox-esr # Update via APT
```

Part C — The FOSS Ecosystem (Units 3 & 4)

Dependencies — What Firefox Is Built On

Library / Component	Role in Firefox
Gecko (Mozilla, MPL 2.0)	Firefox's own rendering engine — parses HTML, CSS, and renders web pages
SpiderMonkey (Mozilla, MPL)	Firefox's JavaScript engine — executes JS on every web page
NSS / NSPR (Mozilla, MPL)	Network Security Services — handles TLS/SSL, certificates, cryptography
SQLite (public domain)	Stores browsing history, bookmarks, cookies, and form data locally
libpng / libjpeg / libwebp	Open-source image decoding libraries for rendering web images
D-Bus / GTK3 (LGPL)	Linux desktop integration — system notifications, file chooser dialogs
Rust (MIT / Apache 2.0)	Mozilla invented Rust; Firefox uses it for memory-safe components (CSS engine Stylo)

What Firefox Has Enabled

Firefox's impact on the web stretches far beyond its user base. Mozilla has created and driven the standardisation of technologies that all browsers now implement: CSS animations, WebGL (3D graphics in the browser), WebRTC (real-time video and voice communication), IndexedDB (client-side databases), and the Web Audio API. These are not proprietary Firefox technologies; they are open web standards created by Mozilla and now implemented by everyone else.

Mozilla invented the Rust programming language. Mozilla created Rust as a tool for writing memory-safe components of Firefox (the CSS engine, the media decoder), but Rust has now become one of the world's most beloved programming languages. Rust is now used by Microsoft, Google, Amazon, and the Linux kernel project for security-critical code. The existence of Rust is a direct result of the creation of Firefox: a browser project that spawned a world-used programming language.

The Mozilla Developer Network (MDN Web Docs), now a collaboration between Mozilla, Google, Microsoft, and Samsung, is the world's best source for web development standards documentation. Mozilla created MDN Web Docs and makes it freely available to every web developer on the planet.

Firefox in the LAMP Stack and Web

In the context of the LAMP stack, Firefox is a client, i.e., a browser through which users can access their Linux, Apache, MySQL, and PHP-based applications. The Firefox developer tools, or "DevTools," are an integral part of the web development environment. They enable developers to examine their HTML and CSS, debug JavaScript, monitor network requests, and analyze performance. They can also be used for simulating mobile browsers. The Firefox DevTools are completely open source, and their influence can be seen in other browser's developer tools as well, except for Firefox itself.

Firefox is also used for server administration. A system administrator using control panels like cPanel, Webmin, or phpMyAdmin through a browser is dependent on Firefox's security model, including certificate management, enforcement of the "same origin" policy, and prevention of cross-site scripting. The open-source nature of a browser means that security can be reviewed.

Community and Governance

Firefox is managed by the Mozilla Foundation, a non-profit organization registered in the US. The source code is hosted on github.com/mozilla/gecko-dev and the Mozilla Phabricator instance at phabricator.services.mozilla.com. The community comprises paid Mozilla staff, volunteer contributors from around the world, and partner developers from organizations like Google and Microsoft, working together on web standards. Mozilla organizes an annual Mozilla Festival, also called MozFest, a celebration of internet health, open source software, and digital rights. There are local communities of Firefox contributors in hundreds of cities worldwide.

Part D — Open Source vs Proprietary (Critical Analysis)

Firefox vs Google Chrome — Comparison Table

Dimension	Firefox (Open Source — MPL 2.0)	Google Chrome (Proprietary — Freeware)
Cost	Completely free. No licensing fees. No hidden costs.	Free to download. Hidden cost: user data collected for advertising.
Security	Fully open source — any researcher can audit the code. Mozilla publishes CVEs publicly.	Built on open-source Chromium, but Chrome itself adds proprietary components. Full audit not possible.
Privacy	Enhanced Tracking Protection blocks third-party trackers by default. No advertising revenue model.	Google's business model depends on advertising. Chrome collects significant telemetry and usage data.
Support	Community forums, MDN documentation, Mozilla support site.	Google support pages and community forums. Enterprise support via Google Workspace.
Freedom to modify	Full freedom under MPL 2.0 — fork, modify, redistribute.	Chromium is open source, but Chrome adds proprietary APIs and DRM components that cannot be modified.
Community control	Mozilla Foundation-governed, nonprofit mission, community contributors.	Google-governed. Direction follows Google's commercial and advertising priorities.
Overall verdict	Recommended for privacy, security auditability, and open web values.	Faster and more compatible with Google services; recommended only where Chrome-specific features are required.

Conclusion of Comparison

Firefox is the right choice for users and organizations who value privacy, security auditability, and open web principles. Firefox is a nonprofit, so their product decisions are not influenced by serving advertisers. Firefox's source is open, so their security claims can be verified rather than taken on faith. Firefox is the right choice for a developer, a journalist, a government agency, or anyone who needs a browser that serves them, not an advertising revenue model for a corporation. The honest truth is, however, that currently, Chrome is ahead of Firefox in many areas from a technical and compatibility perspective. The V8 JavaScript engine in Chrome is currently faster than Firefox's SpiderMonkey. Many web applications are currently tested in Chrome. However, Firefox's gap with Chrome is closing rapidly with each new version of Firefox, and Firefox beats Chrome in all privacy-sensitive benchmarks. Firefox is a browser that I would recommend for use, and it is a browser for which I would contribute. The Mozilla project has a detailed guide for new contributors, and contributions to Firefox's DevTools, localization, and accessibility are actively encouraged.

5. Shell Script Documentation

All five scripts are tested on Kali Linux and submitted as .sh files in the GitHub repository. Add screenshots of each script running where indicated.

Script 1 — System Identity Report

Displays a formatted system welcome screen showing kernel, distro, user, uptime, date, OS licence, and installed Firefox version. Concepts: variables, echo, command substitution \$(), formatted output.

```
#!/bin/bash
# Script 1: System Identity Report
# Author: Khushraj Rai | Registration No: 24BHI10012 | Course: Open Source Software
# Description: Displays a formatted system welcome screen with OS and user info

STUDENT_NAME="[]"
SOFTWARE_CHOICE="Firefox"

# Gather system information using command substitution
KERNEL=$(uname -r)
DISTRO=$(cat /etc/os-release | grep PRETTY_NAME | cut -d= -f2 | tr -d "'")
USER_NAME=$(whoami)
HOME_DIR=$HOME
UPTIME=$(uptime -p)
CURRENT_DATE=$(date '+%d %B %Y, %H:%M')
FF_VERSION=$(firefox-esr --version 2>/dev/null || echo 'Not installed')

# Display formatted output
echo "=====
echo "    OPEN SOURCE AUDIT — SYSTEM IDENTITY REPORT    "
echo "=====
echo "Student      : $STUDENT_NAME"
echo "Software      : $SOFTWARE_CHOICE"
echo "-----"
echo "Kernel        : $KERNEL"
echo "Distribution: $DISTRO"
echo "User          : $USER_NAME"
echo "Home Dir      : $HOME_DIR"
echo "Uptime        : $UPTIME"
echo "Date/Time     : $CURRENT_DATE"
echo "Firefox       : $FF_VERSION"
echo "OS Licence    : GNU General Public License v2 (GPL v2) — Kali Linux"
echo "=====
```

```

script1_system_identity.sh script2_log_analyzer.sh
khush@Khushraj:/mnt/c/Users/khush/college-project/oss-audit-24BHI10012$ chmod +x script1_system_identity.sh
./script1_system_identity.sh
=====
OPEN SOURCE AUDIT - SYSTEM IDENTITY REPORT
=====

Student      : Khushraj Rai (24BHI10012)
Software     : Firefox
=====
Distro       : Ubuntu 24.04.4 LTS
Kernel      : 6.6.87.2-microsoft-standard-WSL2
User        : khush
Home Dir    : /home/khush
Uptime      : up 3 hours, 5 minutes
Date        : Monday, 30 March 2026
Time        : 16:22:06
Firefox     : Firefox not found
=====
OS License   : The Linux kernel is licensed under GPL v2.
               You have the freedom to run, study, modify,
               and distribute this software freely.

Firefox License: Firefox is released under the Mozilla
                Public License 2.0 (MPL 2.0).
                It is a weak copyleft license - modified
                files must stay open source, but it can
                be combined with proprietary code. This
                makes it more flexible than GPL.
=====
khush@Khushraj:/mnt/c/Users/khush/college-project/oss-audit-24BHI10012$ |

```

Script 2 — FOSS Package Inspector

Checks if Firefox ESR is installed, retrieves version and package info, and uses a case statement to print a philosophy note. Concepts: if-then-else, case statement, dpkg, grep.

```

#!/bin/bash
# Script 2: FOSS Package Inspector
# Author: Khushraj Rai | Registration No: 24BHI10012
# Description: Checks package installation and prints an open-source philosophy note

PACKAGE="firefox-esr"

# Check installation using dpkg (Debian/Kali)
if dpkg -l $PACKAGE &>/dev/null; then
    echo "Package      : $PACKAGE is INSTALLED."
    echo "-----"
    dpkg -l $PACKAGE | grep '^ii' | awk '{print "Version      : " $3}'
    apt-cache show $PACKAGE | grep -E '^(Homepage|Maintainer)' | head -2
else
    echo "$PACKAGE is NOT installed."
    echo "Install with: sudo apt install firefox-esr"
fi

echo ""
echo "--- Open Source Philosophy Note ---"

# Case statement: philosophy note per package name
case $PACKAGE in
    firefox-esr|firefox)
        echo "Firefox: The browser that refused to let one company own the web."
        echo "Born from Netscape's defeat, kept alive by millions who chose freedom." ;;
    git)
        echo "Git: Born from proprietary revocation. Freedom by necessity." ;;
    python3)
        echo "Python: A language shaped entirely by its community, not a
        corporation." ;;
)

```

```

    apache2)
        echo "Apache: The web server that made the open internet possible for
everyone." ;;
    vlc)
        echo "VLC: Written by students – proof that open tools outlive corporations." ;;
    *)
        echo "This is free and open-source software. Study its source." ;;
Esac

```

```

khush@Khushraj:/mnt/c/Users/khush/college-project/oss-audit-24BHI10012$ chmod +x script2_package_inspector.sh
./script2_package_inspector.sh
=====
FOSS PACKAGE INSPECTOR – firefox
=====

[🐞] firefox is INSTALLED on this system.

--- Package Details ---
dpkg-query: package 'firefox' is not installed and no information is available
Use dpkg --info (= dpkg-deb --info) to examine archive files.

--- Installed Version ---
./script2_package_inspector.sh: line 58: firefox: command not found

--- Firefox Executable Location ---

--- Open Source Philosophy Note ---
Firefox: When Internet Explorer dominated 95% of the
browser market in 2002, a group of Mozilla developers
feared the web was becoming a closed, proprietary
platform. They built Firefox to prove that a nonprofit,
community-driven browser could compete – and win.
Today Firefox defends the open web against big tech.
=====
khush@Khushraj:/mnt/c/Users/khush/college-project/oss-audit-24BHI10012$ |

```

Script 3 — Disk and Permission Auditor

Loops through key system directories, reports size, permissions, owner, group, and checks Firefox's profile directory. Concepts: for loop, arrays, du, ls -ld, awk, cut.

```

#!/bin/bash
# Script 3: Disk and Permission Auditor
# Author: Khushraj Rai | Registration No: 24BHI10012
# Description: Audits permissions and disk usage for important system directories

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "-----"
printf "%-18s %-8s %-22s\n" "Directory" "Size" "Perms / Owner / Group"
echo "-----"

# Loop through each directory in the array
for DIR in "${DIRS[@]}; do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld "$DIR" | awk '{print $1, $3, $4}')
        SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)
        printf "%-18s %-8s %-22s\n" "$DIR" "${SIZE:-N/A}" "$PERMS"
    else
        echo "$DIR does not exist on this system."
    fi
done

# Check Firefox's user profile directory
echo ""
echo "--- Firefox Profile Directory Check ---"
FF_PROFILE="$HOME/.mozilla/firefox"

```

```

if [ -d "$FF_PROFILE" ]; then
    echo "Firefox profile found at: $FF_PROFILE"
    PERMS=$(ls -ld "$FF_PROFILE" | awk '{print $1, $3, $4}')
    SIZE=$(du -sh "$FF_PROFILE" 2>/dev/null | cut -f1)
    echo "Permissions : $PERMS"
    echo "Profile size: ${SIZE:-N/A}"
else
    echo "No Firefox profile found. Launch Firefox once to create it."
fi

```

```

khush@Khushraj:/mnt/c/Users/khush/college-project/oss-audit-24BHI10012$ chmod +x script3_disk_permission_auditor.sh
./script3_disk_permission_auditor.sh
=====
DISK AND PERMISSION AUDITOR - SYSTEM REPORT
=====
Directory                Perms (Owner:Group)    Size
-----
/etc                      drwxr-xr-x (root:root) 4.3M
/var/log                  drwxrwxr-x (root:syslog) 41M
/home                     drwxr-xr-x (root:root) 24K
/usr/bin                  drwxr-xr-x (root:root) 95M
/tmp                      drwxrwxrwt (root:root) 24K
=====

--- Firefox-Specific Installation Audit ---

[!] Firefox binary not found in PATH.

[!] Firefox profile directory not found at /home/khush/.mozilla/firefox

=====
khush@Khushraj:/mnt/c/Users/khush/college-project/oss-audit-24BHI10012$ |

```

Script 4 — Log File Analyzer

Reads a log file line by line, counts keyword occurrences, and prints the last 5 matching lines. Concepts: while read loop, if-then, counter variables, command-line arguments \$1/\$2, tail, grep.

```

#!/bin/bash
# Script 4: Log File Analyzer
# Usage: ./log_analyzer.sh /var/log/syslog [keyword]
# Author: Khushraj Rai | Registration No: 24BHI10012

LOGFILE=$1
KEYWORD=${2:-"error"} # Default keyword is 'error' if not provided
COUNT=0

# Validate that a log file argument was provided
if [ -z "$LOGFILE" ]; then
    echo "Usage: $0 <logfile> [keyword]"; exit 1
fi

# Check file exists and is a regular file
if [ ! -f "$LOGFILE" ]; then
    echo "Error: $LOGFILE not found."; exit 1
fi

# Check if the file is empty
if [ ! -s "$LOGFILE" ]; then
    echo "Warning: $LOGFILE is empty. Nothing to analyse."; exit 0
fi

echo "Analysing : $LOGFILE"
echo "Keyword   : $KEYWORD"
echo "-----"

# Read the file line by line using a while-read loop
while IFS= read -r LINE; do
    # If the current line contains the keyword (case-insensitive), increment counter

```

```

        if echo "$LINE" | grep -iq "$KEYWORD"; then
            COUNT=$((COUNT + 1))    # Arithmetic expansion to increment
        fi
    done < "$LOGFILE"

    echo "Keyword '$KEYWORD' found $COUNT time(s) in $LOGFILE"
    echo ""

    # Display last 5 matching lines if any were found
    if [ $COUNT -gt 0 ]; then
        echo "Last 5 matching lines:"
        grep -i "$KEYWORD" "$LOGFILE" | tail -5
    fi

```

```

=====
khush@Khushraj:/mnt/c/Users/khush/college-project/oss-audit-24BHI10012$ chmod +x script4_log_analyzer.sh
./script4_log_analyzer.sh
=====
LOG FILE ANALYZER - FOSS AUDIT
=====
[!] No log file specified.
Usage: ./script4_log_analyzer.sh /path/to/logfile [keyword]

Common log files to try:
    /var/log/syslog      (general system log)
    /var/log/auth.log    (authentication events)
    /var/log/kern.log     (kernel messages)
khush@Khushraj:/mnt/c/Users/khush/college-project/oss-audit-24BHI10012$ |

```

Script 5 — Open Source Manifesto Generator

Interactively asks 3 questions and writes a personalised open-source philosophy statement to a .txt file. Concepts: read, string concatenation, file output > and >>, date, alias concept.

```

#!/bin/bash
# Script 5: Open Source Manifesto Generator
# Author: Khushraj Rai | Registration No: 24BHI10012
# Note on aliases: 'alias ff="firefox-esr"' creates a shortcut for Firefox.
# Aliases are defined in ~/.bashrc for persistence across terminal sessions.

echo "=====
echo "  OPEN SOURCE MANIFESTO GENERATOR  "
echo "=====

# Gather user input interactively using read -p
read -p "1. One open-source tool you use every day: " TOOL
read -p "2. In one word, what does freedom mean to you? " FREEDOM
read -p "3. One thing you would build and share freely: " BUILD

DATE=$(date '+%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"    # Filename uses current username

# Write manifesto to file - > creates/overwrites, >> appends each new line
echo "Open Source Manifesto - $DATE" > "$OUTPUT"
echo "-----" >> "$OUTPUT"
echo "" >> "$OUTPUT"
echo "Every day I rely on $TOOL - a tool built openly and shared freely." >> "$OUTPUT"
echo "To me, freedom means $FREEDOM - and that is what open source represents." >>
"$OUTPUT"
echo "It is the right to learn from and build upon the work of those who came
before." >> "$OUTPUT"
echo "Firefox was built because someone refused to let one company control the web." >>
"$OUTPUT"
echo "One day I will build $BUILD and release it openly," >> "$OUTPUT"
echo "so someone I will never meet can stand on my shoulders, as I stood on others." >>
"$OUTPUT"

```



```
echo "That is the open source promise. That is my commitment." >> "$OUTPUT"

echo ""
echo "Manifesto saved to: $OUTPUT"
cat "$OUTPUT"
```

```
khush@Khushraj:/mnt/c/Users/khush/college-project/oss-audit-24BHI10012$ chmod +x script5_manifesto_generator.sh
./script5_manifesto_generator.sh
=====
OPEN SOURCE MANIFESTO GENERATOR - VITyarthi
=====

You chose Firefox - a browser built by a nonprofit to
keep the web open, free, and accountable to its users.

Answer three questions to generate your manifesto.
-----

1. Name one open-source tool you use every day: Mozilla Firefox
2. In one word, what does 'freedom' mean to you? Choice
3. Name one thing you would build and share freely: An educational coding platform
-----

Generating your Firefox open-source manifesto...

[📄] Manifesto saved to: manifesto_khush.txt
=====
MY OPEN SOURCE MANIFESTO - FIREFOX EDITION
Generated on : 30 March 2026
Author      : khush
=====

I chose Firefox as the subject of my open-source audit
because it represents something rare: a piece of software
built not to generate profit, but to protect people.

Every day, I use Mozilla Firefox - a tool that exists because
someone believed in building openly and sharing freely.
To me, freedom means Choice. Firefox embodies that
through the Mozilla Public License 2.0 - ensuring that
any changes to its code must also remain open source,
so the web stays accountable to its users, not corporations.

Inspired by this, I would build An educational coding platform and release it
freely - because the open web only survives if people
like us choose to contribute back to it.

Open source is not just a license. It is a promise:
that the tools we depend on will never be turned against us.

- khush, 30 March 2026
=====
=====
khush@Khushraj:/mnt/c/Users/khush/college-project/oss-audit-24BHI10012$ |
```


6. Conclusion

This audit of Firefox has been an audit of software as an act of resistance. Firefox was not created in response to a product vision or a business opportunity; it was created in response to a particular moment in time when one company was about to monopolize access to the internet, and a community of developers decided that that was not acceptable. The Mozilla Public License version 2.0 that it's distributed under is a considered compromise between protecting the engine itself from being quietly taken over and allowing commercial products to be created around it.

The lesson of the existence of Firefox is that a web browser is not a neutral object; it is the window through which most of humanity reads the news, accesses education, communicates with one another, and participates in public life. The people who create that window, and the interests that they serve in creating that window, are of enormous importance. The existence of Firefox is a reminder that a non-profit community can create world-class software, create industry-wide standards, create new programming languages, and defend user rights against trillion-dollar competitors. Understanding that is part of understanding what open source is actually for.

7. References

- Mozilla Foundation. About Mozilla: <https://www.mozilla.org/en-US/about/>
- Mozilla. Firefox Source Code Repository: <https://github.com/mozilla/gecko-dev>
- Mozilla Public Licence 2.0 full text: <https://www.mozilla.org/en-US/MPL/2.0/>
- GNU Project. The Free Software Definition: <https://gnu.org/philosophy/free-sw.html>
- Open Source Initiative. The Open Source Definition: <https://opensource.org/osd>
- Raymond, E.S. The Cathedral and the Bazaar: <https://catb.org/~esr/writings/cathedral-bazaar>
- MDN Web Docs (Mozilla Developer Network): <https://developer.mozilla.org>
- SPDX Licence List — MPL 2.0: <https://spdx.org/licenses/MPL-2.0.html>
- Choose a Licence — MPL 2.0: <https://choosealicense.com/licenses/mpl-2.0/>
- Shotts, W. The Linux Command Line: <https://linuxcommand.org/tlcl.php>
- GNU Bash Manual: <https://gnu.org/software/bash/manual>
- The Rust Programming Language (Mozilla): <https://www.rust-lang.org>